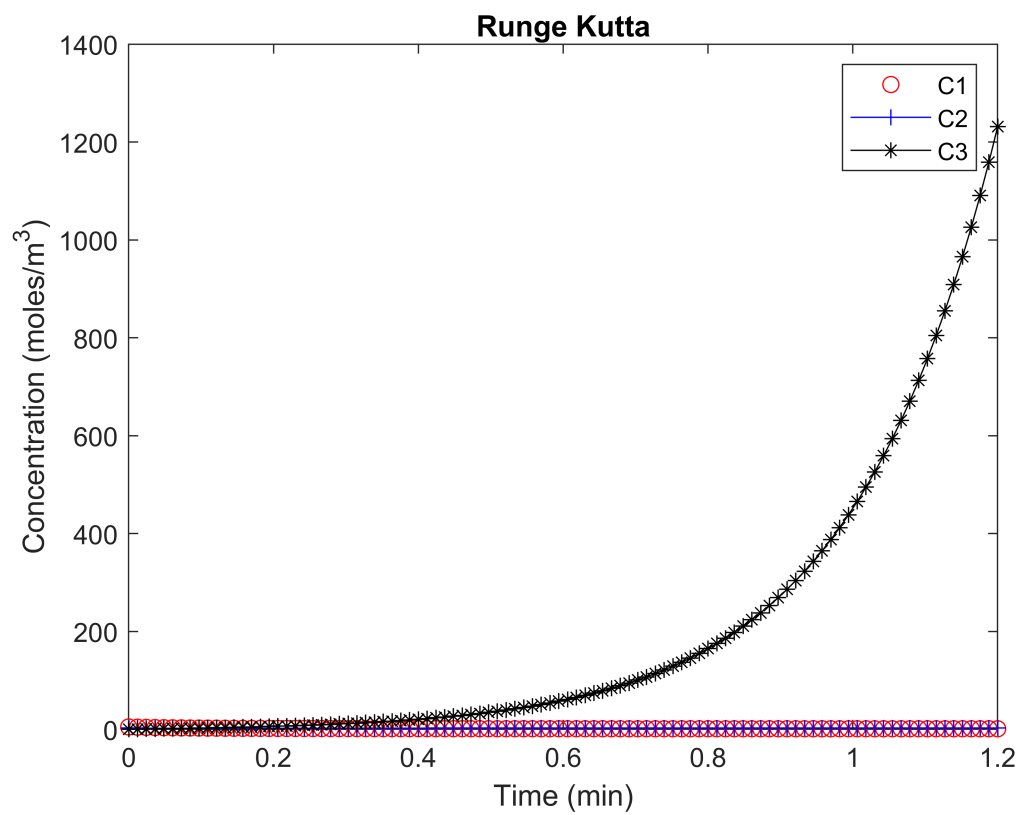
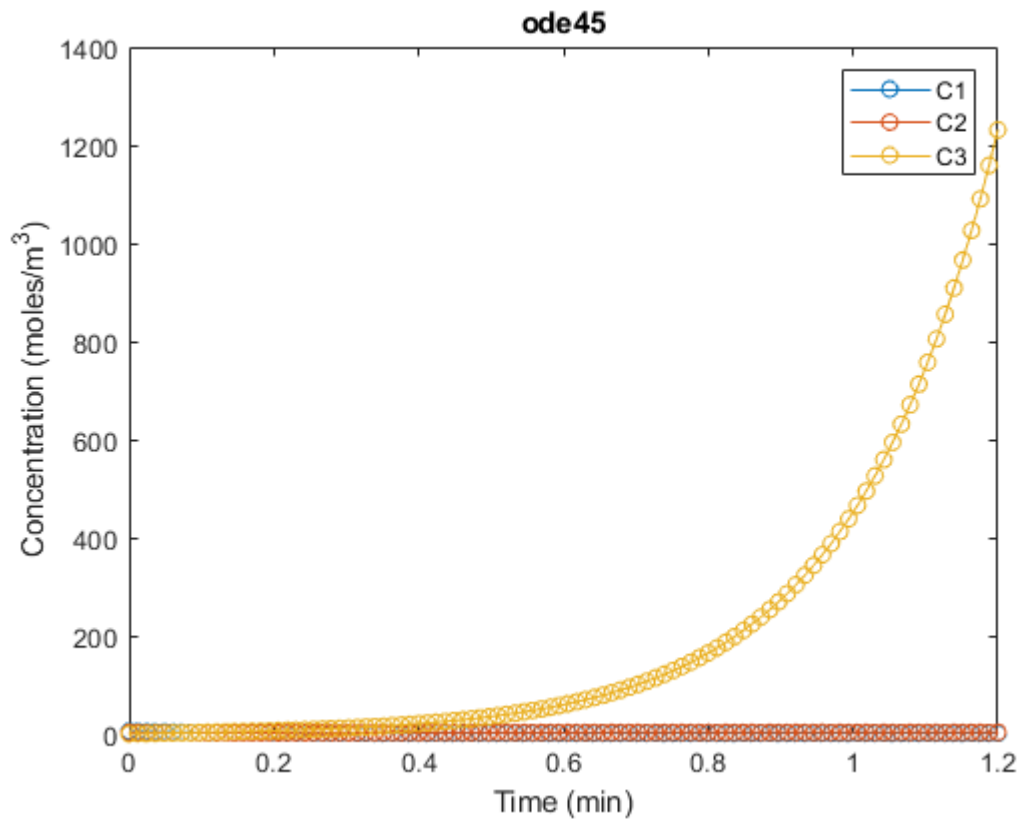


Exercise 2

```
t=linspace(0,1.2);  
C=[5; 3; 0];  
g=@odesyst2;  
[result]=runge_kutta(g,t,C);  
C1 = result(1,:);  
C2 = result(2,:);  
C3 = result(3,:);  
plot(t,C1,'ro',t,C2,'-b+',t,C3,'-k*')  
title('Runge Kutta')  
xlabel('Time (min)')  
ylabel('Concentration (moles/m^3)')  
legend('C1','C2','C3');
```



```
ode45(g,t,C)  
title('ode45')  
xlabel('Time (min)')  
ylabel('Concentration (moles/m^3)')  
legend('C1','C2','C3');
```



ode45 and runge_kutta give the same result.

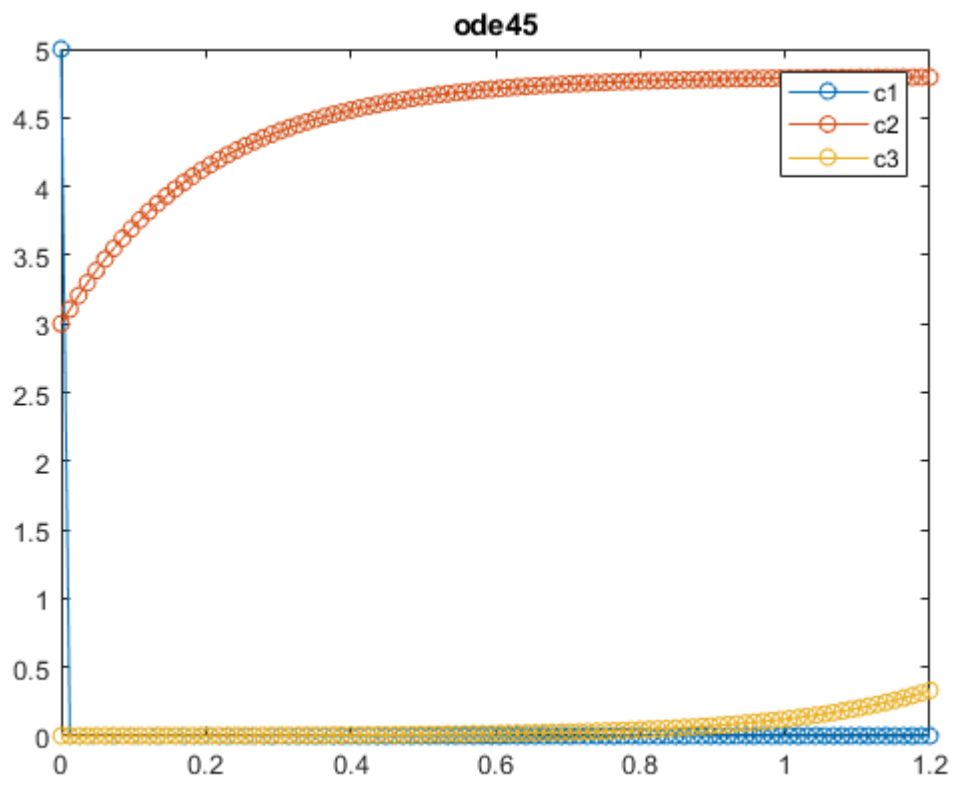
Solving the ode system using ode45 and ode23s with $\alpha_1 = 100000$, $\alpha_2 = 1$, $\alpha_3 = 10$.

```
[c]=runge_kutta(@odesyst,t,C)
```

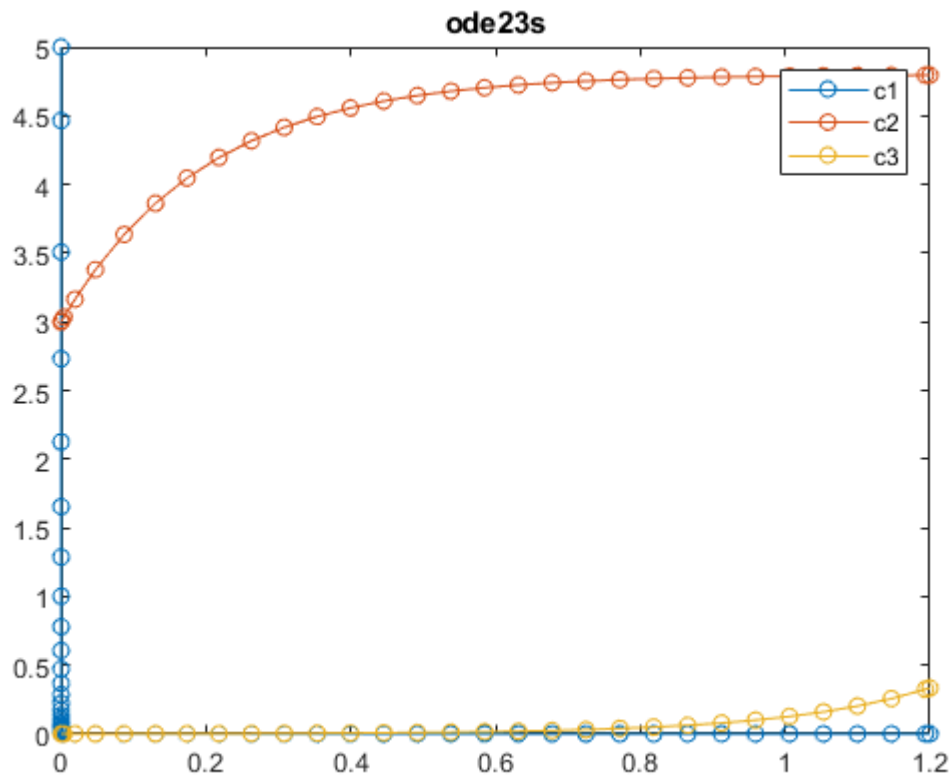
```
c = 3×100
1022 ×
    0.0000    -2.1851    -Inf         NaN         NaN         NaN         NaN         NaN ...
    0.0000    -0.0000    -Inf         NaN         NaN         NaN         NaN         NaN
    0         0.0002     Inf         NaN         NaN         NaN         NaN         NaN
```

Using Rungw Kutta does not work for n=4. For n=100 it will also not work because we have a stiff ode system.

```
t=linspace(0,1.2);
C=[5; 3; 0];
ode45(@odesyst,t,C)
title('ode45')
legend('c1','c2','c3')
```



```
ode23s(@odesyst,[0 1.2],C)
title('ode23s')
legend('c1','c2','c3')
```



It took 6 seconds for ode45 to run. In contrast, ode23s took 0.41 seconds.

```
function [dcdt] = odesyst(t,c)
c01=3.2;c02=4.8;Q=10; V=2;
dcdt=zeros(3,1);
dcdt(1)=(Q/V)*(c01-c(1))-100000*c(1)*c(2);
dcdt(2)=(Q/V)*(c02-c(2))-1*c(1)*c(2);
dcdt(3)=(Q/V)*c(3)+10*c(1)*c(2);
end
function [dcdt] = odesyst2(t,c)
c01=3.2;c02=4.8;Q=10; V=2;
dcdt=zeros(3,1);
dcdt(1)=(Q/V)*(c01-c(1))-2.5*c(1)*c(2);
dcdt(2)=(Q/V)*(c02-c(2))-2.5*c(1)*c(2);
dcdt(3)=(Q/V)*c(3)+2.5*c(1)*c(2);
end
```

```
function [y] = runge_kutta(df,x,y0)
m = length(y0);
n = length(x);
% y is a matrix of size 'mxn'
y=zeros(m,n);
y(:,1)=y0;
```

```

for i=1:n-1
    h = x(i+1)-x(i);
    xi=x(i);
    yi=y(:,i);
    k1 = feval(df,xi,yi);
    k2 = feval(df,xi+0.5*h,yi+0.5*k1*h);
    k3 = feval(df,xi+0.5*h,yi+0.5*k2*h);
    k4 = feval(df,xi+h,yi+k3*h);
    y(:,i+1) = y(:,i) + h*(k1+2*k2+2*k3+k4)/6;
end
end

```